

Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and Deepfake Image Detection

R26-IT-011

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Presentation Outline

- introduction
- System Architecture Overview
- The Problem
- The Proposed Solution
- Individual Contributions
  - Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition
  - Age and Gender Detection using Face-API
  - Facial condition detection
  - Deep fake image detection
- Commercialization
- Charts

Video Description



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Introduction

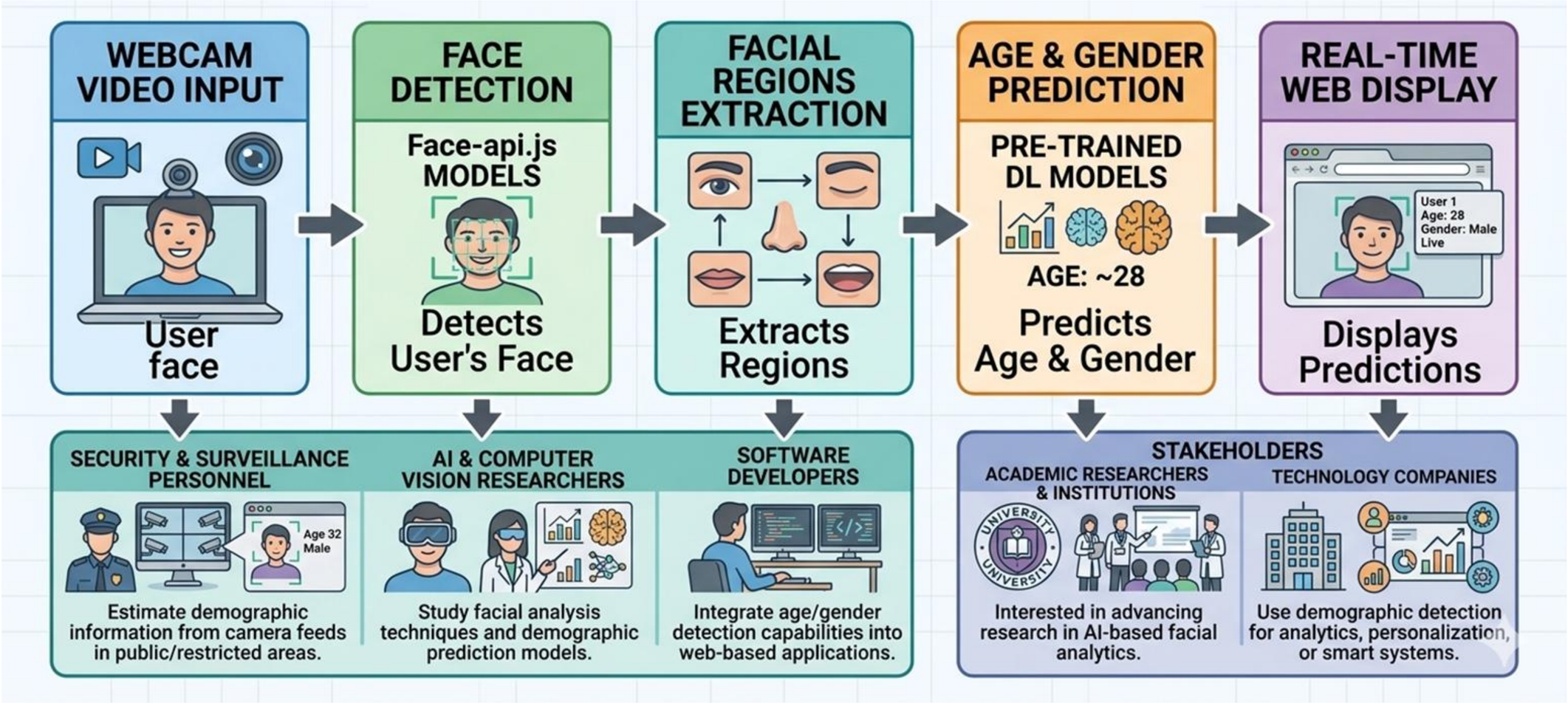
- ***The Concept***
  - *A comprehensive facial analysis framework designed specifically for the South Asian demographic.*

- ***Core Technology***
  - *A fully browser-based solution using Face-api.js, MobileNet V2, and TensorFlow.js.*

- ***Key Goal***
  - *To bridge the gap between Western-centric AI models and the unique skin profiles and facial structures of our region.*

- ***Privacy-First***
  - *All processing happens on the user's device (Client-side), ensuring biometric data security.*

Users & Stakeholders



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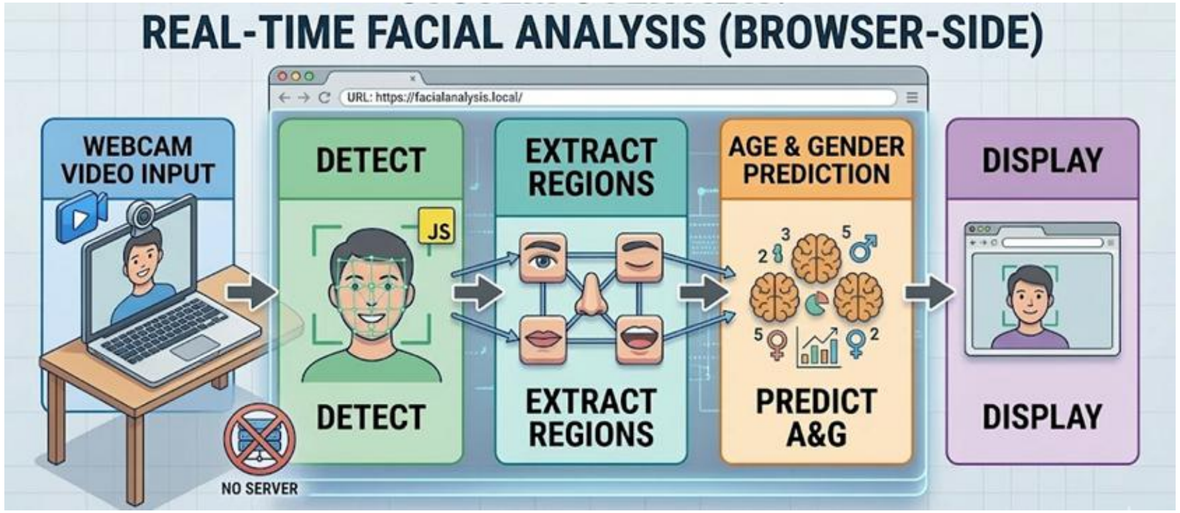
Impact of the problem

- Privacy Risks
- Trust and Ethical Concerns
- Limited Accessibility to facial analysis technologies
- Reduced Efficiency in systems

Existing Solutions & GAP

- Most solutions require cloud-based processing or expensive SDK licenses.
- Many systems depend on high-performance hardware or server infrastructure.(Visage SDK)
- Limited availability of lightweight browser-based solutions accessible to general users.

Proposed Solution



- Uses browser-based deep learning models through Face-api.js and TensorFlow.js.
- Performs real-time face detection and demographic prediction using webcam input.
- Provides a privacy-friendly approach where facial data is processed locally on the user's device.
- Can run on standard computers and web browsers without specialized hardware.

Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition



- ***Contribution to the overall solution***
  - **Performance Benchmarking:** Measuring real-time metrics to ensure the system is fast enough for practical use.
  - **Bias Identification:** Specifically identifying where Face-api.js fails when processing South Asian facial structures compared to Western ones.
  - **System Optimization:** Providing data-driven feedback to the team to improve the model's detection stability.

- ***Novelty of the Component***
  - **Cross-Cultural Accuracy Analysis:** One of the few studies to specifically benchmark Face-api.js performance on South Asian phenotypes.
  - **Quantifying "Accuracy Parity":** Introducing a mathematical way (Accuracy Parity Index) to measure if the AI is "fair" to our people.

Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition

• *Problem Definition*

- ***The Problem:*** *Generic facial AI tools are mostly trained on Western faces and optimized for bright lighting.*

• *Main Objective*

To scientifically measure and evaluate how well Face-api.js recognizes emotions on South Asian faces compared to industry standards.

• *Sub - Objectives*

- ***Testing for Fairness (Bias):*** *Compare accuracy differences between South Asian faces and Western faces (Mitigating bias).*
- ***Low-Light Test:*** *Measure how much accuracy drops in dark environments.*

Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition

- ***Research Methodology***
  - ***Data Categorization***
  - ***Accuracy & Bias Analysis***
  - ***Low-Light Robustness Evaluation***

- ***Latest Technologies***

- **Lightweight Neural Networks (SSD MobileNet V1):** A fast and efficient AI model designed to work on everyday devices like laptops and phones.
- **In-Browser Processing (TensorFlow.js):** The technology that lets the AI run locally on the user's computer, making it fast and private.
- **Automated Enhancement (Zero-DCE):** A simple method to brighten dark images so the AI can "see" faces more clearly in low light.

Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition

- ***Validation and Evaluation Plan***
  - **Accuracy Benchmarking:** Comparing the AI's emotion predictions against the SA-FED dataset to calculate the success rate.
  - **Cross-Cultural Comparison:** Testing the system on both South Asian and Western face groups to find and measure the "Accuracy Gap."
  - **Environmental Stress Test:** Testing the system in different light levels (Bright vs. Dim) to see how robust the detection is.

- ***Data Availability, Security, and Ethics***
  - **Data Sources:** Utilizing the SA-FED dataset and other public research datasets for benchmarking and analysis.
  - **Privacy-First Security:** Local Inference. Since the AI runs in the browser, no facial images are sent to or stored on any external server.

  Ethical Compliance: * All data is used purely for research purposes.
    - Ensuring Algorithmic Fairness for South Asians by identifying and reducing detection bias.

Evaluating the Accuracy and Performance of Face-API.js Emotion Recognition

- ***System Overview of the Individual Component***
  - ***Role:*** *Acts as the Quality & Performance Monitor for the entire CAFI framework.*
  - ***Input:*** *Receives processed facial data from the "Illumination Restoration" (Zero-DCE) and "Emotion Recognition" (Face-api.js) modules.*
  - ***Output:*** *Provides Performance Reports and Bias Metrics to show if the system is fair for South Asians.*

- ***Dependencies and Common Components***

SA-FED Dataset: *Needed to test the system on South Asian faces.*

Face-api.js Weights: *The trained AI models required to run the detection.*

Zero-DCE Model: *Required to check how well the system works after brightening the images.*

Web Browser Platform: *The shared environment where the entire system runs locally.*

Age and Gender Detection using Face-API



Provides real-time age and gender prediction for detected faces, adding demographic analysis to the overall facial recognition system.

Problem Definition

- **Most facial analysis systems rely on cloud/server-based processing**
- **Network latency affects real-time performance**
- **Need for a lightweight browser-based AI solution**

Research Objective

- **Develop a web-based system to detect faces and estimate age and gender in real time**

Novelty

- **AI processing directly in the browser**
- **Compatible with Caucasian face identification(Sri Lanka)**

Age and Gender Detection using Face-API

Research Methodology

- Uses concepts from Artificial Intelligence and Computer Vision
- Facial feature extraction using Face-Api Js
- Real-time image processing from webcam video frames

Technologies

- Face-api.js for face detection and demographic prediction
- TensorFlow.js for running deep learning models in the browser
- TypeScript for web-based implementation

Validation and Evaluation

- Evaluate performance under different lighting conditions
- Measure processing speed and real-time responsiveness

Age and Gender Detection using Face-API

Data Availability, Security & Ethics

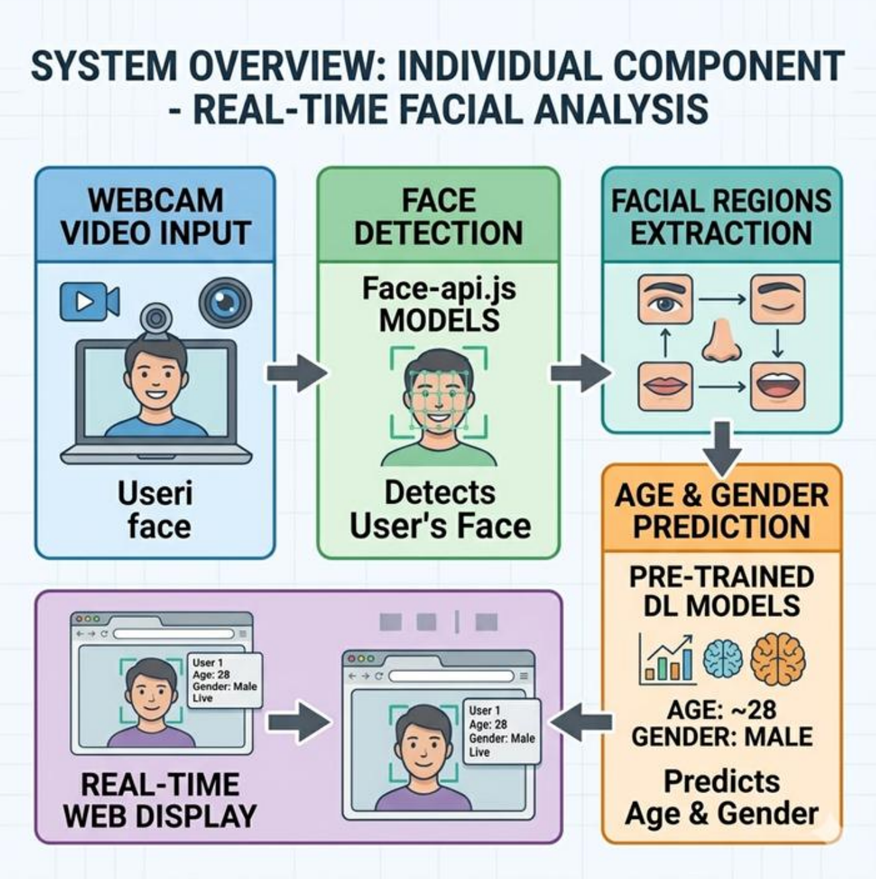
- Uses pre-trained models and publicly available datasets
- Facial images processed locally in the browser
- No permanent storage of user facial data

Research Gap

- Limited research focuses on real-time browser-based AI solutions for facial demographic analysis.
- Many models are designed for high-performance GPU environments, making them difficult to deploy on lightweight devices.

IMDB-WIKI Dataset
Adience Dataset
UTKFace Dataset

Age and Gender Detection using Face-API



Dependencies & Components

- Face-api.js – Face detection and demographic prediction
- TensorFlow.js – Running deep learning models in the browser
- Webcam API – Capturing real-time video input

Facial Skin Condition Detection



Contribution to the overall solution

- *Detects specific skin condition as visual point on the image,providing a clear diagnostic overlay for the system.*
- *optimizing the overall system to perform accurately on srilanka skin profile reducing demographic bias.*

Novelty of your component

- *Enhance the diagnostic layer specifically trained for Srilankan skin types and condition.*
- *Use styleGAN2-ADA to generate hight fidelity synthetic data for rare skin condition.*
- *YOLOv11 for accurate lesion detection and classification.*

Facial Skin Condition Detection

Problem Definition

- *Most facial anlysis model are trained on european skin, causing issue when applied to south asian skin tones.*
- *Static facial features like wrinkles, acnes, and more can be misidentified reducing system accuracy.*
- *Humidity and srilanka can affect fine skin feature detection.*

Main Objective

To develop a hight precision context aware physilogical diagnosticlayer for detecting facial condition in Srilanka skin profile.

Novelty

- *Facial analysis models trained on European skin often misperform on Srilankan skin.*
- *Wrinkles, acne and others are often misidentified, lowering facial analusis accuracy.*
- *Humidity and strong sunlight in Srilanka can impair fine skin feature detection.*

Sub Objective

- *To implement a deep learning model to detect skin lesions and visualize them on the face images.*
- *To optimize the model for Srilankan skin type and condition to improve detection reliability.*
- *To utilize StyleGAN2-ADA to generate hight fidelity synthetic images for Srilanka skin pathologies overcoming dataset scarity.*

Facial Skin Condition Detection

Research Methodology

- *Data Collection & South Asian Dataset Preparation*
- *Perform image preprocessing and augmentation*
- *Train deep learning models for face condition detection*
- *Edge Optimization & Quantization*

Latest Technology

- *YOLOv11 Architecture*
- *StyleGAN2-ADA*
- *Train deep learning models for face condition detection*

Facial Skin Condition Detection

Validation, Data & Ethics.....

Validation and Evaluation

- *Mean Average Precision (mAP).*
- *Inference Latency Testing.*
- *Evaluation detection accuracy using(precision, Recall).*

Data Availability, Security & Ethics

- *Data Availability.*
- *Informed Consent & Anonymization.*
- *Secure storage of images in cloud database.*

Facial Skin Condition Detection

Implementation Details.....

System Overview

- *Input & Pre processing.*
- *AI Inference (YOLOv11).*
- *Visual Point Mapping & Reporting.*

System Overview

- *StyleGAN2-ADA*
- *Deep Learning models(YOLOv11).*
- *image processing pipline.*

Deepfake Image Detection



- ***Contribution to the overall solution***
  - ***Primary Security Engine:*** *Develops a high-precision framework to distinguish authentic Sri Lankan faces from hyper-realistic AI-generated imagery.*
  - ***Data Integrity Gatekeeper:*** *Verifies the authenticity of facial data before it enters the emotion recognition pipeline, ensuring results are based on real human expressions.*
- ***Novelty of The component***
  - ***Bridging the Security Gap: Face-API.js specializes in tracking and emotions but lacks the capability to verify if a face is real or synthetic.***
  - ***The framework specifically fine-tuned for Sri Lankan skin textures and cultural facial features using a specialized dataset.***

Deepfake Image Detection

- ***Problem Definition & Objectives :***

- ***The Problem:*** *Demographic Bias: Existing models suffer from a "Hyperrealism Bias" towards Western features, leading to lower accuracy for local demographics.*

- ***The Problem:*** *Small Data Problem: A lack of high-fidelity synthetic and authentic facial datasets for regional groups like Sri Lankans.*

- ***Main Objective :***

  Develop and validate a demographically unbiased deepfake detection framework optimized for Sri Lankan faces

- ***Sub Objectives :***
  - *Build a balanced dataset of real and AI-generated Sri Lankan faces.*
  - *Ensure the system is accurate and unbiased for local Sri Lankan demographics.*

Deepfake Image Detection



- ***Research Methodology***
  - ***Data gathering***
  - ***Pre-processing***
  - ***Accuracy & Bias Analysis***
  - ***Model Selection & Evaluation***

- ***Latest Technologies***

  - ***Vision Transformer (ViT) or Swin-Base:*** *Modern transformer-based architectures that, unlike traditional CNNs, can capture global structural dependencies and microscopic skin artifacts essential for identifying hyper-realistic deepfakes.*
  - ***Synthetic Synthesis (The Gap Solution):*** *Utilizing StyleGAN2-ADA to generate "fake" counterparts that perfectly match the Sri Lankan demographic profile.*
  - ***Error Level Analysis (ELA):*** *A forensic pre-processing method used to detect pixel-level compression anomalies and non-uniform artifacts often found in deepfakes.*

Deepfake Image Detection

- ***Validation and Evaluation Plan***
  - ***Accuracy Benchmarking:*** *Comparing the AI's classification (Real vs. Fake) against standardized datasets such as DiffusionFace, Celeb-DF, and FaceForensics++ to calculate the overall success rate.*
  - ***Phenotype Analysis:*** *The system is tested using both Western faces and South Asian/Sri Lankan faces.*
  - ***Digital Artifact Robustness:*** *Measures how well the compression system performs when reducing the quality of images through social media (WhatsApp, FB), which is common in Sri Lanka.*

- ***Data Availability, Security, and Ethics***

Data Sources: *Utilizing a combination of public research datasets (DiffusionFace, Celeb-DF) and a localized dataset of Sri Lankan faces to ensure representative benchmarking.*

Privacy-First Security: *Since the AI runs directly in the browser using Face-api.js and Vision Transformers, all facial processing is performed on-device; no images are sent to or stored on any external servers.*

Ethical Compliance: * All data is used purely for research purposes.

- *Ensuring fairness for South Asian individuals by identifying and minimizing detection bias across different phenotypes.*
- *Anonymizing all research results to ensure that individual identities are never recorded or compromised.*

Deepfake Image Detection

- **System Overview of the Individual Component**
  - ***Role:*** *Acts as the Deepfake Verification & Forensic Diagnostic Layer for the overall framework.*
  - ***Input:*** *Uploaded facial images or real-time camera streams.*
  - ***Output:*** *Authenticity Probability Score .*

- **Dependencies and Common Components**

StyleGAN2-ADA Synthesis Engine: *Required to generate high-fidelity synthetic samples to balance the local dataset.*

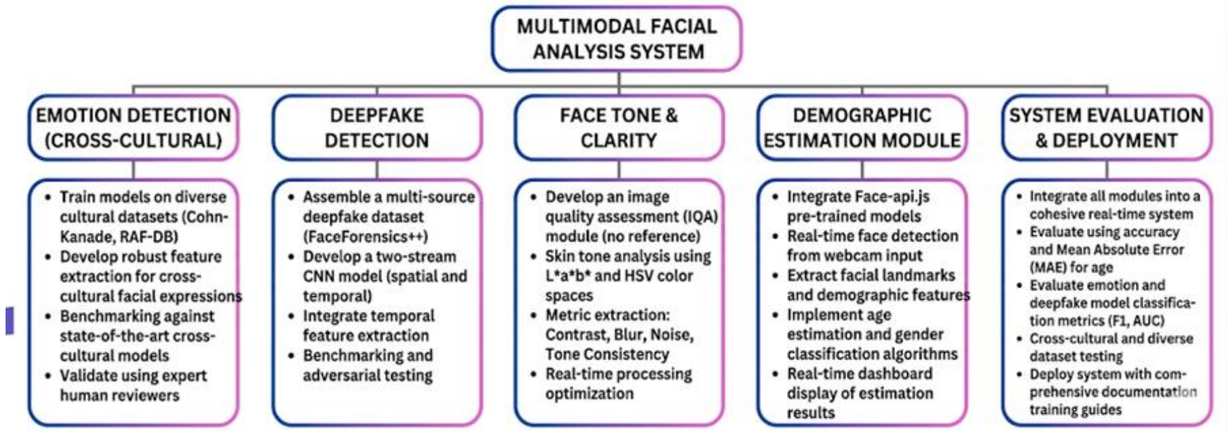
Web Browser Platform: *The shared environment where the entire system runs locally.*

Commercialization

GANT CHART

Phase	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct
Literature Review & Benchmarking								
AI Model Development								
Image Preprocessing & Testing								
UI/UX Design								
App Development								
Face-API testing and Lightweight model testing								
Final Validation and PWA activation								

WORK BREAKDOWN STRUCTURE



Commercialization

ESTIMATED BUDGET

Item	Estimated Cost (LKR)	Purpose
Software Tools (Face-api.js, TensorFlow.js, VS Code, Git)	0	Open-source tools used for implementing face detection, age estimation, and gender classification.
Development Environment (Laptop / Desktop with Webcam)	0	Personal or university-provided device used for coding, testing, and running the web application.
Internet Access	2,000	Required for downloading libraries, models, and accessing documentation during development.
Testing Environment	0	Testing performed using available webcams and browsers on existing computers.
Documentation and Report Preparation	1,000	Printing, formatting, and preparing the final research report and supporting documents.
Miscellaneous / Contingency	2,000	Small buffer for unexpected development or testing expenses.

Commercialization

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GitHub repository: https://github.com/justadudewhohacks/face-api.js
— Provides pre-trained models for age and gender estimation that can run directly in web browsers.

Smilkov, D., et al. (2019).
TensorFlow.js: Machine Learning for the Web and Beyond.
arXiv preprint.
— Framework for running deep learning models like CNNs in the browser, enabling age and gender detection on client-side.

Levi, G., & Hassner, T. (2015).
Age and Gender Classification Using Deep Learning.
CVPR Workshops.
— Focuses on model architectures and evaluation metrics specifically for demographic prediction.

Thank You

Q&As



